

3. Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0

∴ The condition that for each pivot column, all elements below the pivot position is 0 is True.

So, the matrix $\begin{bmatrix} 1 & -3 & h \\ 0 & 1 & -5 \\ 0 & 0 & 2h+7 \end{bmatrix}$ is in echelon form.

There is a row $[0 \ 0 \ 2h+7]$. By Theorem 2, the system can only have a solution when $2h+7=0$, else the system is inconsistent.

∴ $h = -7/2$
∴ $y = \begin{bmatrix} -7/2 \\ -5 \\ -3 \end{bmatrix}$ is in the plane generated by $v_1 = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$ and $v_2 = \begin{bmatrix} -3 \\ 1 \\ 8 \end{bmatrix}$. □

2) Let $u = \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix}$ and $v = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$. Show that $\begin{bmatrix} h \\ k \end{bmatrix}$ is in $\text{Span}\{u, v\}$ for all h and k .

Solution: By Defn. of Span, a vector $\begin{bmatrix} h \\ k \end{bmatrix}$ is in $\text{Span}\{u, v\}$ if $\begin{bmatrix} h \\ k \end{bmatrix}$ is a linear combination of u and v , i.e. $x_1 u + x_2 v = \begin{bmatrix} h \\ k \end{bmatrix}$ for some $x_1, x_2 \in \mathbb{R}$

$$\text{or, } x_1 \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} x_2 = \begin{bmatrix} h \\ k \end{bmatrix}$$

This is equivalent to claiming that the augmented matrix

$\begin{bmatrix} 2 & 2 & h \\ -1 & 1 & k \end{bmatrix}$ has a solution set.

$$\begin{bmatrix} 2 & 2 & h \\ -1 & 1 & k \end{bmatrix} \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ + \frac{1}{2} \times \text{Row 1}}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 2 & 2 & h \\ 0 & 2 & k+h/2 \end{bmatrix}$$

Verification of the matrix in echelon form

1. There are no zero rows in the matrix. ∴ The condition that all non-zero rows must be above all zero rows is vacuously True.